

SYNOPSIS



At 0052 on 6 March 2004, the chemical tanker *Reno*, and the fishing vessel *Ocean Rose*, collided. *Reno* was carrying a cargo of acrylonitrile and was on passage from Teesport to Immingham; *Ocean Rose* was engaged in fishing. A further collision between the two vessels occurred several minutes later when *Reno* was manoeuvring at slow speed to provide assistance. *Ocean Rose* sustained major damage during the first collision, but with the assistance of the coastguard and the Staithes and Whitby lifeboats, was able to reach Whitby under her own power. The damage to *Reno* was superficial.

As a power-driven vessel, *Reno* was required to keep clear of *Ocean Rose*, but did not do so because her OOW had left the bridge and gone to his cabin. The AB lookout saw *Ocean Rose* and, realising that she was potentially a problem, tried to contact the OOW. He was unsuccessful. When the two vessels were very close, the AB lookout took avoiding action by altering course to starboard, and the skipper of *Ocean Rose* altered course to port. This put the vessels on a collision course.

After the collision, the AB lookout alerted both the OOW and the master. *Reno* was manoeuvred towards *Ocean Rose* with the master controlling the ship's speed, and the OOW on the helm. Some of the crew felt the impact and started to check the main engine, cargo, fuel, void and ballast spaces for damage; others, including the chief officer, played no part in subsequent events. As a result, some checks were not as comprehensive or timely as might otherwise have been the case, and tasks such as preparing the lifeboat were never confirmed as being completed.

It is considered that several factors on board *Reno* contributed to this accident, and subsequent events:

- The OOW was not on the bridge.
- Rather than inform the master of the OOW's absence or the proximity of *Ocean Rose*, the AB lookout preferred to take avoiding action himself.
- The master was possibly overloaded or distracted after the first collision.
- The general alarm was not sounded.

Recommendations have been made to the International Chamber of Shipping, and the International Shipping Federation aimed at: reinforcing the requirement for OOWs to remain on the bridge at all times when on duty, via direction, education and training; achieving acceptable levels of response from crews in emergency situations through the provision of realistic drills; and promoting an understanding of the potential effect of nationality and cultural mix on ships' operations and procedures.

1.2 BACKGROUND

Reno

Reno, a 2238grt chemical tanker, was on passage from Teesport to Immingham, carrying a cargo of 2,438.584 tonnes of acrylonitrile (UN number 1093), which poses a pollution and safety hazard. The ship mainly operated in the North Sea and was a regular visitor to Teesport.

Ocean Rose

Ocean Rose, a 17.5m stern trawler from Whitby landed her catch in Whitby during the evening of 4 March and sailed on completion. During 5 March, three tows were conducted without incident, with each tow lasting about 4 hours. The vessel intended to return to port for the weekend on completion the fourth tow.

1.3 NARRATIVE

All times are UTC, all courses are true

1.3.1 *Ocean Rose*

At about 0030, on 6 March 2004, the skipper of *Ocean Rose* went to the wheelhouse to take over the watch in readiness for hauling. The autopilot was engaged with a north-westerly course selected. The vessel was towing her fishing gear into the tidal stream at a speed over the ground of about 2 knots. The skipper switched on the vessel's deck lights. The sidelights, stern light, and green over white trawling lights at her masthead were already illuminated.

A radar contact was seen closing from ahead at a range of about 3 miles. The vessel was also visible, but the skipper could not determine her aspect because of the brightness of her deck lights. By about 0045, the skipper was concerned by the proximity of this contact, and changed to manual steering. When the approaching vessel was about one mile away, the skipper used a hand-held signal lamp to alert her by twice flashing one long followed by two short illuminations (letter D in Morse code) towards her. The vessel continued to close. When within about 2.5 cables, the skipper put the helm hard over to port, he also hurried the deckhands below to prepare to haul. The skipper altered to port because he considered the approaching vessel would just pass clear to starboard and he intended to increase her CPA. As he was towing his fishing gear, he also wanted to optimise the effects of the tidal stream to assist the turn.

When the two deckhands arrived in the wheelhouse, the skipper immediately passed the signal lamp and told them to go forward and get the vessel's name. He also told them to confirm that the fishing and navigation lights were on, which they did.

By the time the two deckhands reached the foredeck, the approaching vessel was within one cable ahead of *Ocean Rose*. With the aid of the signal lamp, the deckhands were able to read the vessel's name, *Reno*, on her starboard bow. While they were doing this, the skipper called the approaching vessel on VHF channel 16 at 0052 and 18 seconds and stated:

I've come hard over for you, steam boat. I don't know what your name is. I'm going to get your name and report you to the coastguard. You are going to hit us here, you are going to hit us

The oncoming vessel suddenly appeared to turn towards *Ocean Rose* and, with collision imminent, the skipper shouted for his crew to hold on. He also disengaged the engine and repeated on VHF radio channel 16 at 0052 and 36 seconds:

You are going to hit us, you are going to hit us

The collision occurred at 0052 and 38 seconds, and was almost head on. *Ocean Rose* was hit on her stem and starboard bow, which caused her to roll heavily to port. The skipper was thrown to the port side of the wheelhouse, and the deckhands fell forward on to the deck. The skipper immediately assessed that *Ocean Rose* had suffered substantial damage, and sent a "Mayday" via VHF radio, and by pressing the distress button on his Inmarsat radio. The damage to the forward part of *Ocean Rose* is shown at **Figure 1**.

Figure 1



Damage to *Ocean Rose* - forward

1.3.2 *Reno*

Reno sailed from Teesport at about 2300 on 5 March. At about 2315, the second officer left his mooring station aft and joined the master and pilot on the bridge to take his watch. The pilot disembarked at 2328, and at about 2345 the master handed over the watch to the second officer. The master completed several administrative tasks on the bridge, and then retired to his cabin. This left the second officer and an AB lookout on the bridge. Course was 110° in autopilot, and speed was between 11.5 and 12 knots. There were major discrepancies between the accounts of the second officer and the AB lookout regarding subsequent events.

The second officer's account

At about 0010, a radar target was detected at about 8 miles on the starboard bow. The radar was operating on the 6-mile range scale in relative motion, and off-centred to the west-north-west. The target's relative track was almost parallel to the ship's heading; the second officer assessed that it was almost stationary and would pass about 1 mile to starboard. This was supported by ARPA, which indicated the target was on a course of 290° at a speed of between 2 to 3 knots, with a CPA of between 0.5nm and 1nm. It was not considered to be a problem. Soon after, the second officer saw the vessel visually, and through binoculars observed her bright deck lights and a green light. The second officer associated the lights he saw with a fishing vessel, but he was not sure if she was engaged in fishing.

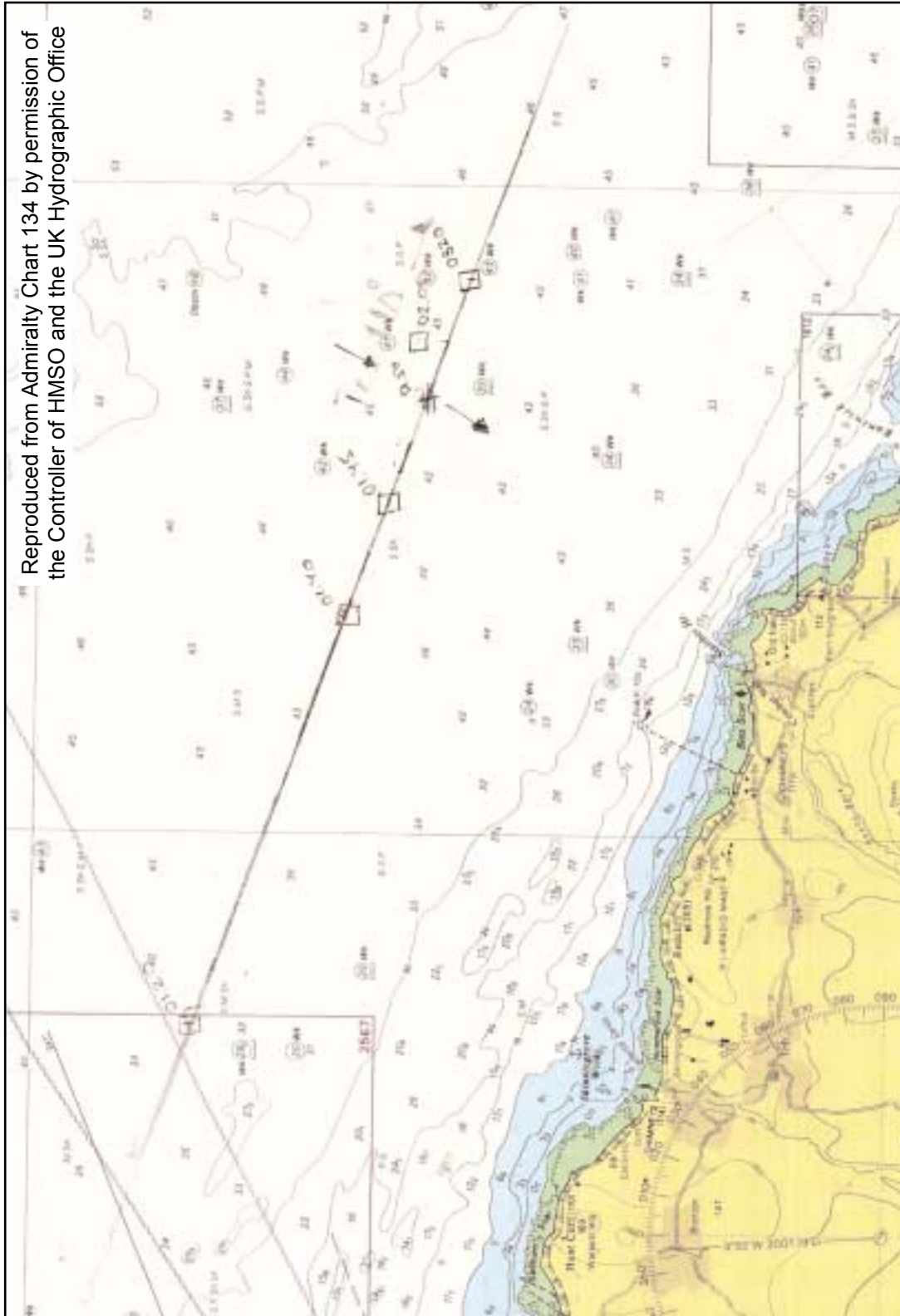
At about 0040, when the fishing vessel was at about 3 miles, her CPA remained between 0.5 and 1 mile. About 4 minutes later, however, when the fishing vessel had closed to about 2 miles, her bearing became steady. The second officer then heard the fishing vessel calling on VHF radio channel 16, and recalls her trying to establish radio contact on several occasions thereafter. The second officer did not respond because he did not want to get involved in a VHF radio conversation, which might have confused the issue. By now, the second officer was mainly monitoring the fishing vessel visually rather than by radar, using fixed points about the bridge to gauge her bearing movement. He did not rely on ARPA information as the system had lost the fishing vessel target on two or three occasions as the vessels closed.

The fishing vessel was now very close and, as the second officer tried to determine her aspect, he saw that the green light he had seen previously was at her masthead. He also assessed that the vessel had altered course to starboard to a more northerly course. Hand steering was selected and 30° of starboard helm was applied to avoid her. As the ship started to turn, the second officer realised the fishing vessel was now altering to port. He also saw the fishing vessel flashing a white signal light in the direction of *Reno*, but did not recognise the flashes as having any particular meaning. The second officer reversed the helm back to port, but the two vessels collided. The AB lookout was

immediately ordered to call the master. After hearing the fishing vessel's "Mayday" on VHF radio channel 16, the second officer applied starboard helm to close her position.

During this period, the second officer plotted fixes on the paper chart at 0040 (GPS), 0045 (radar) and 0050 (GPS). These are shown at **Figure 2**.

Figure 2



The AB lookout's account

After the master left the bridge, the AB maintained a lookout while moving between the port and starboard radar displays. The second officer was working on charts at the chart table, but checked the radar displays periodically. At some point, the second officer informed the AB that he was going down to his cabin to get changed out of his winter overalls that he had been wearing for the mooring operations. He checked the radar displays before leaving the bridge. There were no signs of any vessels, but lights on the shore to starboard were visible.

Later, the AB saw a fishing vessel about 2 miles off the starboard bow. Through binoculars, he saw a green over white light, and a second green light, and assumed that she was engaged in fishing. He also assumed that she would pass down the starboard side. The AB lookout tried to tell the second officer about the fishing vessel, but after receiving no response, realised that the second officer was still absent from the bridge. The AB telephoned the second officer's cabin, and also the mess room and the cargo office. There was no reply, so the AB left the bridge and went to the second officer's cabin on the deck below, and found the door closed. He did not knock on the door or try to open it, but went down another level and shouted for the second officer down the stairway leading to the mess room. Again, there was no response. The AB ran back up the two flights of stairs to the bridge. He estimates that he was away from the bridge for no more than 60 seconds.

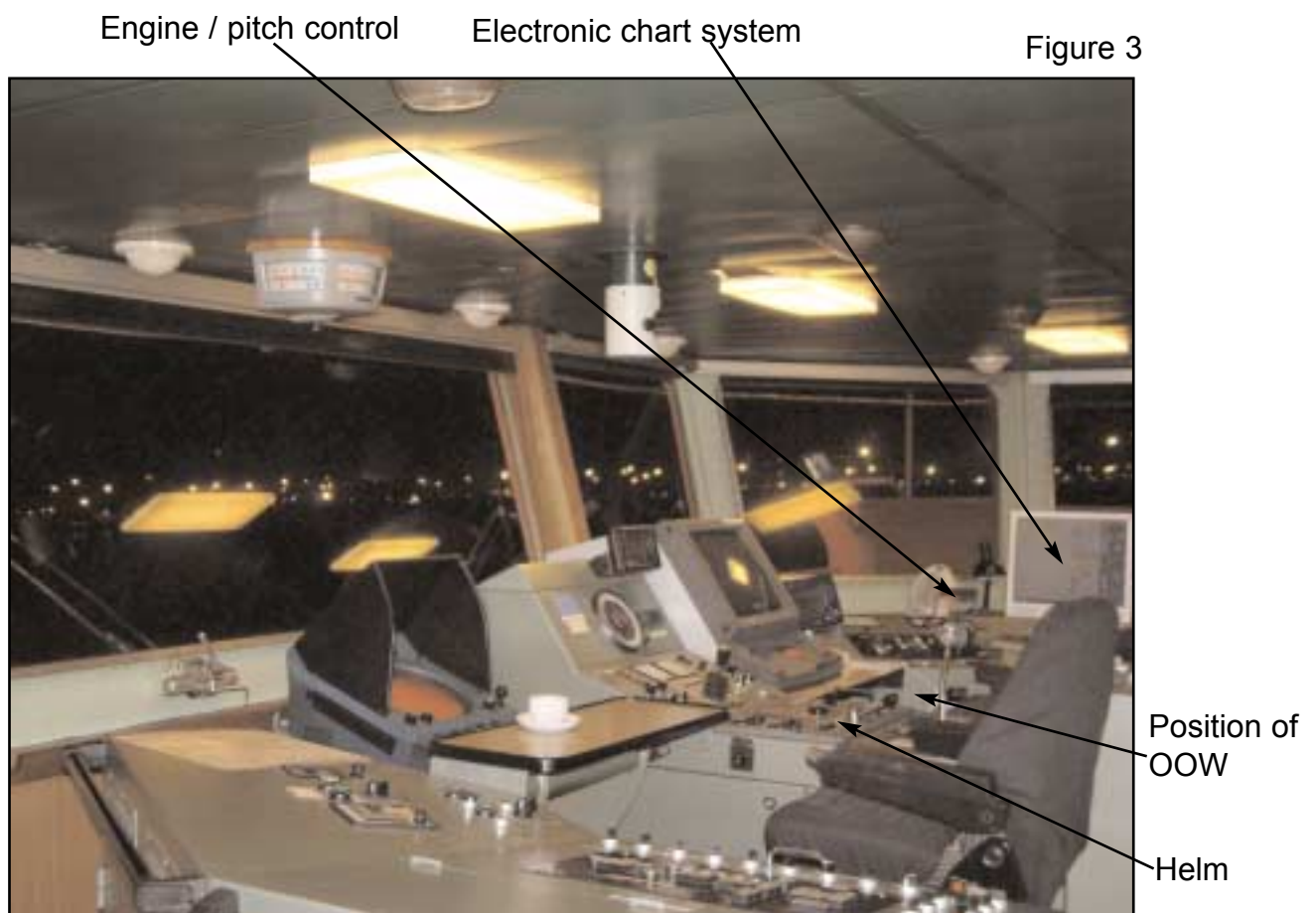
On his return to the bridge, the fishing vessel was about 1 mile away on the starboard bow, and showing the same lights as before. Soon after, however, when the fishing vessel was very close, the AB saw one of the green navigation lights change to red, and thought that she was trying to cross ahead. It was, however, very difficult to determine her aspect because of the brightness of her deck lights. He also saw the fishing vessel flashing a signal light, and heard her calling on VHF radio. To avoid the fishing vessel, the AB adjusted course to starboard using the push button controls on the automatic pilot, and recalls the rudder indicator showing 15°, which was the limit set when steering in this mode. As *Reno* started to turn, the AB saw that the fishing vessel was now altering course to port and, although he tried to stop the turn to starboard with the automatic pilot, the vessels collided.

After the collision, the AB ran to the deck below and opened the second officer's cabin door. The second officer was standing in the toilet area of his cabin; he was fully clothed but had changed his overalls. The AB told him that *Reno* had collided with a fishing vessel and that he must go to the bridge immediately. The AB also alerted the master before he returned to the bridge.

1.4 SUBSEQUENT MANOEUVRING

Reno's master was still up and about in his cabin at the time of the collision, but did not feel the ship manoeuvring. He did feel the collision, however, and was readying himself to go to the bridge when the AB arrived. He went straight to the bridge and asked what had happened. The AB informed him that the ship had been in a collision with a fishing vessel. The second officer did not reply. The master could see a fishing vessel on the starboard quarter, which appeared to be manoeuvring. The pitch control lever was still at position '8', the position it had been in when he had left the bridge earlier. The master ordered the second officer to put it to 'stop'. The vessel's engine control record indicates this action was taken between 0054 and 0055. As the master also considered the vessel to be on her original course of 110°, he also ordered the second officer to apply starboard helm.

Reno manoeuvred towards *Ocean Rose* with the master operating the pitch control lever, and the second officer on the helm (**Figure 3**). The second officer remained standing in front of the starboard radar display, while the master



Reno bridge layout

periodically moved from a position beside the second officer, towards the radio room located on the port side, aft of the bridge. The master tried to call *Ocean Rose* at 0055 and 37 seconds, and 0058 on VHF channel 16. During the first transmission, the master initially used the sister ship name *Ebro*, a vessel he had commanded previously, instead of *Reno*, but immediately rectified his error. *Ocean Rose*'s skipper did not reply to these calls, but called *Reno* at 0059 and 38 seconds when concerned that the chemical tanker was getting too close. The call stated:

Reno this is Ocean Rose, will you keep clear of our stern sir, we have gear overboard, will you keep clear of our stern sir, we have gear overboard, no chance of moving.

Humber Coastguard immediately asked *Reno* if this had been copied, and at 0100 and 1 second, *Reno*'s master stated:

We are close to the fishing vessel and stopped with her.

As the master finished this statement, the skipper of *Ocean Rose* immediately broadcast that *Reno* had hit him a second time, this time on the stern.

Both the master and second officer recall passing very close to *Ocean Rose* during this period, but were not aware of a second collision between the two vessels. Accounts differ regarding whether *Ocean Rose* disappeared from view under *Reno*'s bow as the vessels closed. *Reno*'s master estimated that the vessels came no closer than 1 cable. The crew of *Ocean Rose* recall that *Reno* approached at an angle of about 90° on their starboard side, and clipped the fishing vessel's stern at a right angle. This is reported to have damaged the power block and roller fair lead on her starboard quarter (**Figures 4 and 5**), and pushed the fishing vessel forward.

1.5 ELECTRONIC CHART SYSTEM

Reno was fitted with a Transas ECS. The vessel's recorded track for the period is at **Figure 6**. The following navigational data was retrieved from the system:

Time (UTC)	Position	Heading	Cog	SoG
0051	54 36.433N 000 43.322W	108.2	111	12.1
0052	54 36.364N 000 43.006W	108.2	111	11.9
0053	54 36.290N 000 42.690W	127.5	115	11.7
0054	54 36.144N 000 42.557W	208.5	192	9.2

SECTION 2 - ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 ACTIONS ON BOARD OCEAN ROSE

Ocean Rose was engaged in fishing and was therefore hampered by her fishing gear. As power-driven vessels are required to keep clear by Rule 18 of the Collision Regulations, the skipper's concern about the lack of avoiding action taken by *Reno* as she approached, was understandable. The skipper's actions to alert *Reno* to his concerns, via the signal lamp and VHF radio, started when the vessels were about one mile apart, and therefore allowed ample time for action to be taken to avoid collision.

When the skipper assessed that *Reno* was not taking action to keep clear, but would pass uncomfortably close down his starboard side, the alteration of course to port would have increased the CPA had *Reno* taken no action. An alteration of course to starboard, which would normally have been appropriate however, would have taken *Ocean Rose* across the bows of *Reno*, and with her fishing gear astern, this action would have been potentially hazardous.

After the initial collision, the skipper's action to broadcast a "Mayday", and instruct his crew to launch the liferaft, was timely and sensible. This resulted in the prompt tasking of rescue resources by the coastguard. Consequently, pumps were provided in time to prevent the loss of the vessel.

On finding the water ingress in the engine room, the mate's action was positive in difficult circumstances. Subsequent inspection of the failed strum box nuts indicated that their corrosion was caused by the use of dissimilar metals. This highlights the need to ensure that all through-hull fittings are properly maintained, and that appropriate materials are used.

2.3 RENO - ASSESSMENT OF DIFFERING ACCOUNTS

The accounts of the OOW and AB lookout were fundamentally different. The AB lookout stated that the OOW had been absent from the bridge for some time before the initial collision, whereas the account of the OOW indicated that he had been on the bridge throughout. There are, however, several anomalies in the OOW's account, which cast doubt over its validity.

First, although it was normal practice for ARPA information to be displayed on the ECS, no such data was recorded during the period in question, despite the fact that the OOW stated that the course, speed, and CPA of *Ocean Rose* had been displayed by ARPA.

Second, the OOW recollected *Ocean Rose* calling on VHF radio when at a range of about 2 miles, and several times thereafter before the first collision. The skipper of *Ocean Rose*, however, called only twice. Also, as the first call was made only 20 seconds before the collision, the distance between the vessels must have been less than one cable at that time.

Third, the rationale of plotting three fixes within ten minutes on the paper chart when in a close quarters situation with a fishing vessel, is difficult to comprehend. It is unlikely that such action would have been taken, and therefore the retrospective plotting of these fixes, to disguise the OOW's absence from the bridge during this period, cannot be ruled out.

Finally, had 30° of helm been applied when taking avoiding action, as stated by the OOW, it is likely that this manoeuvre would have been felt by the master seated in his cabin. Because it was not, this action was more likely to have been taken using 15° of helm in automatic pilot, as described by the AB lookout.

In view of these anomalies, along with the OOW's silence - possibly through ignorance - when asked for information by the master, more reliance is placed in the AB lookout's account. It is considered, therefore, that the OOW was not on the bridge at the time of the first collision.

2.4 RENO - ACTIONS OF THE OOW

The presence of an OOW on the bridge at sea is fundamental to ship safety in every respect, including: navigation, the application of the collision regulations, and reaction to on board emergencies. It is a regulatory requirement, which is considered to be so widely understood that its inclusion in company and master's standing orders and instructions is not usually considered necessary.

On this occasion, the OOW reportedly left the bridge to change his overalls. The length of his absence, however, was much longer than the few minutes this should have taken. When he left the bridge, there were no radar contacts of note on the off-centred radar display set to the 6-mile range scale, and the closing speed with *Ocean Rose* was between 14 and 15 knots. It is therefore possible the OOW was away from the bridge for at least 24 minutes. It is not known what he was doing during this period.

The absence of the OOW from the bridge is unacceptable under any circumstances, and the MAIB is concerned that this type of incident is becoming more common. Only recently, the MAIB investigated another accident caused by a bridge watchkeeper leaving the bridge of a dry cargo ship. The vessel collided with a small, single-handed fishing vessel in the North Sea. The master of the dry cargo vessel, who was the duty bridge watchkeeper, was in his cabin at the time of the collision, and never saw the fishing vessel. The fishing vessel was badly damaged.

It is the MAIB's overriding concern that leaving the navigating bridge of vessels unattended for 'short' periods while vessels are at sea is becoming a more accepted practice among seafarers, rather than a strict taboo.

Owners and managers must ensure, through a targeted regime of direction, audit, training and education, that this attitude is strongly discouraged if further accidents of this type are to be avoided in the future.

2.5 RENO - ACTIONS OF THE AB LOOKOUT

Confronted with an approaching fishing vessel, while alone on the bridge, the AB lookout was placed in a very awkward position, and his initial action to contact the OOW by telephone was understandable. When this was unsuccessful, however, his subsequent actions to leave the bridge and find the OOW, and then to return to the bridge and take avoiding action, and not to inform the master, were inappropriate.

The AB's actions can be partly explained by his friendship with the OOW, with whom he had shared watches over several months. It was equally possible that the AB's culture also influenced his judgment, as the experience of the ship managers was that the Filipino ratings it employed were only comfortable working and reporting within a well-defined hierarchy. Within the bridge watch system, the AB lookout would have been aware of his position in relation to the OOW. However, he might have felt uncomfortable or have been worried about reporting the proximity of the fishing vessel and the absence of the OOW, directly to the master.

As the internationalism of shipping increases, a greater mix of nationalities and cultures on board ships is becoming more commonplace. The dynamics of this mixing of cultures impacts on ships' operations in several ways. Some of its effects are positive. Others, however, possibly have a detrimental effect on the values crews place on safe operation. This is despite all mariners being trained to an international standard. Management awareness of the dynamics of the mixing of cultures is essential to ensure crew effectiveness, and to fully utilise the competencies of the individuals concerned. This might require modifications to manning policies, and to on board organisation and procedures.

2.6 RENO - ACTION FOLLOWING THE FIRST COLLISION

Given the absence of the OOW from the bridge when *Reno* collided with *Ocean Rose*, it is understandable that the immediate reactions on board were not as positive as might otherwise have been the case. Having been informed of the collision by the AB lookout when still in his cabin, the OOW went to the bridge, and would have been in the process of orienting himself when the master arrived.

The master would also have required time to familiarise himself with the situation. As soon as he was aware that the ship had been involved in a collision, however, he ordered the pitch control to be put to 'stop', and starboard helm to be applied to turn towards *Ocean Rose*. He also tried to contact the fishing vessel on VHF radio. These actions were positive and well intended. The general alarm, however, was not sounded in accordance with company instructions and the ICS Bridge Management Guide.

This resulted in several adverse consequences. First, the master undertook several important tasks in addition to his overall command and co-ordination of ship reactions. These included the control of the ship's speed, and VHF radio communication. Second, actions expected of the crew by the master, such as the readying of the lifeboat, were not delegated, monitored or confirmed, and were probably not undertaken. Finally, the assessment of damage and the checking of the cargo tanks were not systematic, and relied to a large extent on the initiative of the chief engineer and the pumpman.

Had the general alarm been sounded, all of the crew would have been immediately alerted to the situation. They would have mustered in one location, and a more systematic and co-ordinated response would have been possible. The chief and third officers could certainly have been used to greater effect with regard to the checking of the cargo, the readying of the lifeboat, and assisting the master on the bridge.

There is nothing to be lost by sounding the general alarm during an emergency, and there is much to gain. Fortunately, because the damage to *Reno* was minimal, there was no pollution, the cargo tanks were intact, nobody was injured or missing, and *Ocean Rose* did not require assistance; the adverse consequences of not sounding the general alarm were not severe. This might not always be the case.

2.7 THE SECOND COLLISION

There are conflicting accounts regarding the occurrence of the second collision between the two vessels, which was reported to have happened about 7.5 minutes after the first collision at 0052. *Reno*'s master states that the vessels came no closer than about 1 cable, whereas the crew of *Ocean Rose* state that *Reno* crossed astern at an angle of about 90° on their starboard side, and made contact with the stern roller and power block as she passed.

As the report of the second collision, made by the skipper of *Ocean Rose* on VHF radio, was spontaneous, it was unlikely to have been triggered by anything other than some form of contact. After the first collision, *Ocean Rose* was anchored by her fishing gear from her stern, and it is likely that the tidal stream would have caused her head to lie in a southerly direction. The ECS data shows

that at 0000, *Reno* was adjacent to the position of the first collision, on a heading of 085°. This supports the fact that the vessels were in close proximity, and the description of the angle of *Reno*'s approach given by the crew of *Ocean Rose*.

The resulting damage reported to be from this contact is difficult to confirm with any certainty. Damage to the power block (**Figure 4**) might have been sustained during the first collision. It is also possible that the damage to the roller fair lead bracket (**Figure 5**) might have been caused by the bulbous bow of *Reno* momentarily snagging the fishing warps passing over the fair lead, rather than via direct contact with the structure itself. Although the major damage in the engine room was first seen after the second collision, it is not certain how the fracture of the pipe, and water ingress via the strum box, could have been initiated by this contact.

Regardless of the nature of the contact, it is evident that *Reno*'s master's perception of the proximity of the two vessels was incorrect. This might have been because at the time of the second collision, he was talking on the VHF radio, and might, therefore, have been distracted or overloaded. This is supported by the fact that at 0100, the master stated on VHF that he was stopped, whereas the speed over the ground recorded on the ECS was 5.8 knots. The error regarding the ship's name when using the VHF, and the use of the wrong telephone number when informing the ship managers, were also possible indicators of the stress the master was under.

2.8 MANNING AND TRAINING

Since flagging into the Madeira register in June 2003, the composition of *Reno*'s crew changed from being predominantly Portuguese, to being multinational. Significantly, a need to train and familiarise the newly recruited crew was highlighted 6 months later by the previous master. Assuming a crew is qualified, the fact that it is multinational should not affect how it reacts in an emergency. However, in order to overcome potential difficulties caused by culture and communication, the responses to emergency situations must be pre-planned, automatic, and well drilled. This can only be achieved via effective training.

In this case, the ship managers had arranged bridge resource management training and training seminars for its senior officers, had employed a qualified chemical tanker master for the purpose of crew training, and had required its ships to conduct a programme of drills, including collision drills. It is disconcerting that, despite these positive measures, the master and crew still did not respond to the initial collision in accordance with the laid down procedures.

Effective drills are often difficult to achieve within the constraints of a commercial schedule, a constantly changing crew, and a lack of external assistance. As a result, many on board drills are only conducted in accordance with a drill plan, and are organised and controlled by the master, who therefore cannot

SECTION 3 - CONCLUSIONS

3.1 SAFETY ISSUES

The following safety issues have been identified by the investigation. They are not listed in any priority order.

1. The actions of the skipper of *Ocean Rose* to alert *Reno* to his concerns, via the signal lamp and VHF radio, allowed ample time for action to be taken to avoid collision. [2.2]
2. The alteration of course to port by *Ocean Rose* would have resulted in an increased CPA had *Reno* taken no action. An alteration to starboard would have caused *Ocean Rose* to cross ahead of *Reno*, which, considering her fishing gear astern, would have been potentially hazardous. [2.2]
3. The broadcast of a “Mayday” via VHF radio, by the skipper of *Ocean Rose*, allowed the prompt tasking of rescue resources by the coastguard. As a consequence, pumps were provided to prevent the loss of the vessel. [2.2]
4. The failure through corrosion of the nuts securing the strum box cover in *Ocean Rose*, highlights the need to ensure that all through-hull fittings are properly maintained at all times, and that appropriate materials are used. [2.2]
5. It is considered that the OOW of *Reno* was not on the bridge at the time of the first collision. [2.3]
6. It is possible that *Reno*’s OOW was away from the bridge for at least 24 minutes, but it is not known what he was doing during this period. [2.4]
7. The absence of *Reno*’s OOW from the bridge on this and previous occasions, along with similar instances occurring during accidents reported to the MAIB, highlights that this practice is more common than ship managers and masters are aware of. [2.4]
8. The actions taken by the AB lookout, without informing the master, were inappropriate. [2.5]
9. The Filipino AB lookout might have felt uncomfortable or have been worried about reporting the situation directly to the master. [2.5]
10. Management awareness of the dynamics of the mixing of cultures is essential to ensure crew effectiveness, and to fully utilise the competencies of the individuals concerned. This might require modifications to manning policies, and to on board organisation and procedures. [2.5]

11. After the collision, the general alarm was not sounded on board *Reno* in accordance with company instructions and the ICS Bridge Management Guide. [2.6]
12. Had the general alarm been sounded, the chief and third officers could certainly have been used to greater effect with regard to the checking of the cargo, the readying of the lifeboat, and assisting the master on the bridge. [2.6]
13. It is difficult to confirm the nature of the contact or the damage caused during the second collision. [2.7]
14. At the time of the second collision, it is evident that *Reno*'s master's perception of the proximity of the two vessels and his vessel's speed, was incorrect. This might have been because he was talking on the VHF radio, and might, therefore, have been distracted or overloaded. [2.7]
15. It is disconcerting that, despite the number of positive measures taken by the ship managers with regard to training, the master and crew still did not respond to the initial collision in accordance with the laid down procedures. [2.8]
16. Greater consideration needs to be given on how the effectiveness of on board drills can be achieved, particularly in view of the ever-increasing number of multinational crews. [2.8]
17. It is likely *Reno*'s OOW remained in his cabin during the period he was absent from the bridge, but it is considered unlikely that he fell asleep. [2.9]